

Supervised Machine Learning 3 (EDFB)

Flipped-Class Activity: Big Data — New Tricks for Econometrics

Work in small groups to explore Hal Varian's paper 'Big Data: New Tricks for Econometrics' and recreate some of its key ideas in a Colab notebook using real data and machine learning tools.

What to Do

1. 1. Read & discuss (≈30 min)

- Go through Varian's paper (JEP 2014, link: <https://www.aeaweb.org/articles?id=10.1257/jep.28.2.3>). As you read, note examples where Varian uses supervised learning (e.g. regression, trees, random forests).

- Discuss in your group:

- What problem is he trying to solve?
- How does ML help compared to traditional econometrics?

2. 2. Recreate & explore (≈60–70 min)

- Open a Google Colab notebook.

- Download Varian's replication data from OpenICPSR (<https://www.openicpsr.org/openicpsr/project/113925>).

- Step by step, try to:

1. Load and inspect the data (pandas / head() / describe()).
2. Fit a simple linear regression as a baseline.
3. Fit a machine learning model (e.g., random forest).
4. Compare predictions and discuss differences.

- You can use AI tools for code help — but make sure you understand what's happening!

3. 3. Wrap-up (last 15–20 min)

- Each group summarizes in 2–3 slides or notebook cells:

- What you replicated or learned
- What the ML model added (or didn't add)
- One insight connecting ML to econometrics

Deliverable

A short Colab notebook showing:

- Some replication of Varian's analysis (data, model, results)
- Brief written reflections (in text cells)